# Machine Learning in Python

Last Update: October 16, 2026 ©Maximilien Notz 2026

## Numpy

## Pytorch

### Tensor

Let `t` be a pytorch tensor and the following are from `torch`:

| | |
|---|---|
| `.randn(s)` | Create a random tensor of rank 1 |
| `.ones(s)` | Create a tensor of ones of rank 1 |
| `.zeros(s)` | Create a tensor of zeros of rank 1 |
| `.cat(lst,dim=d)` | Concatenate the tensors in lst along dimension `d` |
| `t.size()` | Return the size of the tensor `t` |
| `t.view(s)` | Reshape the tensor `t` to the shape |

### Deep Learning

All the following are in `torch.nn`:

| | |
|---|---|
| `.linear(m, n)` | Apply a afine linear transformation object with `m` inputs and `n` outputs. |
| `.conv1d(m, n, k)` | Apply 1D convolution with `m` input chanels and `n` output chanels with a kernel `k`. |
| `.conv2d(m, n, k)` | Apply 2D convolution with `m` input chanels and `n` output chanels with a kernel `k`. |
| `.conv3d(m, n, k)` | Apply 3D convolution with `m` input chanels and `n` output chanels with a kernel `k`. |

`print(obj,sep='|',end='.',file=sys.stdout,flush=False)`
Prints `obj` to the specified `file` with a separator `sep` between objects and an ending character `end` at the end

### Activation Function

All `Pytorch` are not performed in place unless we add the argument `inplace=False`.

| | |
|---|---|
| `.Sigmoid()` | $\sigma(x) = \frac{1}{1+e^{-x}}$ |
| `.ReLU()` | $\text{ReLU}(x) = \max(0, x)$ |
| `.LeakyReLU(negative_slope=m)` | $\text{LeakyReLU}(x) = \begin{cases} x & x > 0 \\ mx & x \leq 0 \end{cases}$ |
| `.Tanh()` | $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ |

### Regularization

| | |
|---|---|
| `.Dropout(p=P)` | Dropout regularization with probability `P` |
| `.BatchNorm()` | Batch normalization |

## Scikit Learn

## Base Models

### XGBoost

### LightGBM

### ROCKET

### MINIROCKET